1- Qualitative vs Quantitative Data

• Quantitative data : where data tends to be Subjective impression rather than numeric values , (description without numbers)

•Quantitative data : data that contains numbers ,collected by counting or measuring ,has 2 types (discrete & continuous)

- discrete data : only certain values

- continuous data : almost any value

2-Types of data

- A- Nominal scale data :
  - Qualitative or Categorical
  - Order doesn't matter
- B- Ordinal scale data:
  - Used for ranking
  - Order matters
  - Can't measure difference
- C- Interval scale data:
  - Order matters
  - Difference can be measured ,but can't be a ratio
  - No true "0" starting point

- D- Ratio scale data:
    - Order matters
    - Difference are measured and Ratios
    - Contains “0” starting point (ex. Grades)

3-Hypothesis

• Hypothesis testing : a proses that uses sample statistics to test a claim about the value of a population parameter

- Why do we even need it ?
- -For Data-driven decision-making
- 1-Formulate Hypothesis -> 2-Find the right test -> 3-ececute the test -> 4-Make a decision

•To make a hypothesis:

- we need a null hypothesis $H0$ & alternative hypothesis Ha
    - $H0$: is a statistical hypothesis that contains a statement of equality <=, >= or =
    - Ha : statement of inequality , complementary to $H0$

-The concept of null hypothesis is like :

(innocent till proven guilty) we make a claim and try to prove otherwise.

- To test a hypothesis we have 2 types of tests:

1- Z-test : if normal population and σ is known

Or not normal population, n≥30

2- T-test : if normal population and σ is unknown, n<30

- Types of Error in tests:

1- α : occurs if the $H0$ is rejected when it's True.
2- β : occurs if the $H0$ is not rejected when it's False.

3- Confidence Interval

- -It's the range between lower and upper limit from a sample statistics that mostly contains the true population parameter .
- -ex. Confidence level = 95% -> then ,Z=1.96 (from z table)
- -Formula : C = $\bar{x} \pm z \times \frac{std}{\sqrt{n}}$

,Where (x̄:mean , z value for confidence level , n: sample size)

- -the confidence level can be calculated for other parameters , as it only shows in which range the parameter lies with a certain probability.

4- Regression Analysis

-It's a method for modeling relationships between variables, enables to infer or predict a variable (dependent or output) depending on one or more variables (independent or input)

- Why regression is important?

1- Measure the influence of one or more variables on another variable
2- Prediction of one variable by one or more other variables

- Types of Regression:

1-Simple Linear Regression: uses only one variable for prediction

2-Multiple Linear Regression: uses more than one variable for prediction

3-Logistic Regression : for nominal or categorical variables.

(ex. Binary logistic: has 2 values 0 or 1, yes or no … etc)